

EmoMime: Augmenting Social Behavior and Self-Expression via Wearable Robotic Limbs

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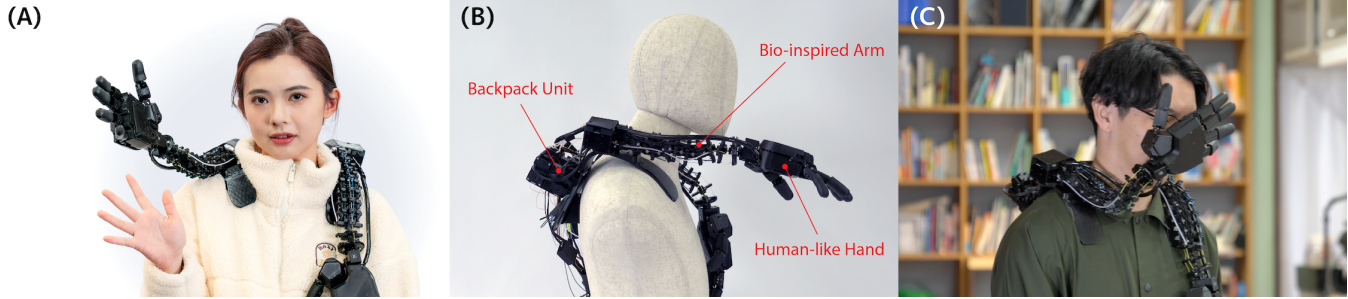


Figure 1: EmoMime: a wearable robotic system designed to augment social behavior and self-expression. (A) Social Interaction: A robotic hand, positioned near the face, dynamically amplifies social behaviors like waving. **(B) System Configuration:** Human-like hands, bio-inspired flexible arms, and a backpack unit for face proximity. **(C) Self-Expression:** The soft mechanism enables safe face contact.

Abstract

While hand-over-face gestures serve as critical multimodal channels for conveying emotions, existing Supernumerary Robotic Limbs (SRLs) have been designed primarily for external task execution using rigid structures. This task-oriented design poses significant challenges for delicate interactions near the face. We propose EmoMime, a novel wearable robotic system designed to augment social behavior and self-expression within the peripersonal space. By integrating a human-like hand for semantic expressiveness with a bio-inspired flexible arm for passive safety, the system resolves the trade-off between safety and expression. This unique configuration ensures physical and psychological safety, enabling natural non-verbal communication in close proximity to the face. In the demonstration, the user can experience augmented social presence

through scenarios such as expressive amplification, social mediation, and self-regulation, showcasing a future where technology physically embodies our internal states.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computer systems organization** → **Robotics**.

Keywords

Supernumerary Robotic Limbs, Wearable Robotics, Human Robot Interaction, Hand-shaped Interface



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1 Introduction

In face-to-face communication, hand-over-face gestures, such as touching the chin while thinking or covering the mouth in surprise, serve as critical multimodal channels to convey emotions and internal states [Mahmoud et al. 2011]. Wearable robots have the potential to augment these near-face gestures through physical motion and presence.

However, existing supernumerary robotic limbs [Inami et al. 2022; Sasaki et al. 2017], which often feature rigid arm structures designed to expand outward workspaces or bodily expression, are not suitable for near-face applications. For example, most existing robotic arms are anchored laterally on the waist or back. This configuration physically separates the hands from the face, the center of social attention, making it difficult for gestures made by the end-effector to function effectively as social signals. Furthermore, trajectories approaching the face from the outside involve psychological pressure and collision risks, making them unsuitable for delicate interactions in the immediate facial region.

We propose *EmoMime*, a wearable robot designed to augment social behavior and self-expression. *EmoMime* realizes interactions within the user’s peripersonal space, specifically around the face. Through three core functions: (1) expressive amplification, (2) social mediation, and (3) self-regulation, *EmoMime* pioneers a new horizon of Human-Robot Interaction within the user’s peripersonal space.

2 Design Concept of EmoMime

EmoMime specializes in interaction proximal to the face, with the objective of augmenting not just physical capabilities but also social expressiveness. Drawing inspiration from the morphological adaptability and flexibility demonstrated in Orochi [Al-Sada et al. 2019], we introduce two design strategies to achieve safe and rich expression within the peripersonal space.

While most wearable arms are laterally mounted to maximize workspace [Inami et al. 2022; Sasaki et al. 2017], *EmoMime* mounts the arm inside the shoulder width, closer to the neck. This “inner positioning” aligns the limbs closer to the user’s centerline. Since gaze naturally gathers at the face, this alignment effectively draws the observer’s attention to social signals. Furthermore, keeping the limbs within the body’s silhouette improves social acceptance and ensures safety in crowded environments.

Robotic interactions near the face present a trade-off between “semantic expressiveness” and “physical safety.” To resolve this, we propose a “Soft-Rigid Hybrid Structure.” A rigid end-effector with high degrees of freedom (DoF) is employed to accurately convey social signals (such as peace signs or pointing) with clear shapes and human-like scale. Meanwhile, a soft manipulator arm provides passive flexibility, ensuring safety during contact or proximity to the face.

3 Implementation

To realize the *EmoMime* concept, we developed a prototype wearable robotic system consisting of four elements: (1) Human-like Robotic Hand, (2) Bio-inspired Flexible Robotic Arm, (3) Neck-mounted Backpack Unit, and (4) Control Software.

3.1 Human-like Robotic Hand

To achieve human-sized expressiveness, we optimized the mechanism of our previous work, Handoid [Shimobayashi et al. 2025]. We retained the original 6 DoF for finger flexion/extension and stable grasping (including thumb base movement), while adding 2 DoF for finger spreading (index-middle finger, middle-ring-little finger). This total of 8 DoF enables diverse social expressions beyond simple grasping (Fig. 2A). Furthermore, a Time-of-Flight (ToF) distance sensor is embedded in the palm to ensure safety during contact.

3.2 Bio-inspired Flexible Robotic Arm

We developed a flexible arm based on the bio-inspired tendon-driven soft robot structure [Nakano et al. 2023]. This mechanism ensures intrinsic safety through passive compliance. The arm has a total of 6 DoF: 4 DoF for the flexible arm sections used to reach, 1 DoF for the shoulder mount for rotation, and 1 DoF for wrist rotation to enhance hand posture. Its 25 cm length enables a natural reach from the chest to the face, ensuring a smooth and safe approach from below that enhances both psychological and physical safety.

3.3 Neck-mounted Backpack Unit

To reduce the burden of long-term wear, heavy components such as servo motors for arm drive and controllers are consolidated in a backpack unit worn on the back. The total weight of the wearable system is 2.2 kg (excluding the power supply unit). It can also accommodate batteries and computing modules. An IMU for measuring body posture is built-in.

3.4 Control Software

The software processes multimodal inputs and triggers context-aware motions. For example, it can instantly react to specific words or changes in voice pitch to perform body gestures. The system architecture supports future integration of Multimodal Large Language Models (MLLMs); adding a camera to the backpack unit would enable the system to visually understand social contexts and behave autonomously.

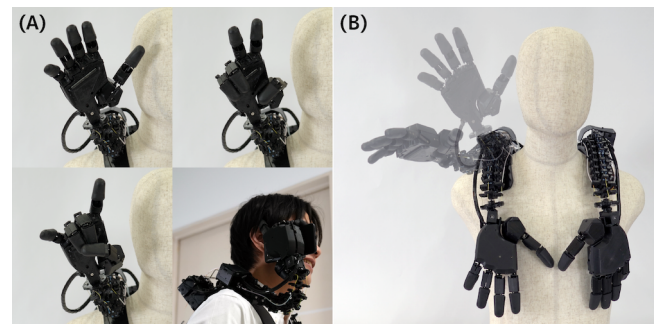


Figure 2: Prototype of EmoMime. (A) The robotic hand with eight Degrees of Freedom enables diverse social gestures. (B) The bio-inspired arm provides a wide range of motion, smoothly lifting the hand from the chest to the face region.

4 User Experience

At SIGGRAPH '26, attendees are invited to wear *EmoMime* and experience an augmented social body that bridges physical assistance and social expression through three interaction scenarios:

Expressive Amplification Even for shy users, *EmoMime* reacts to quiet voices (e.g., “Nice!”) by performing a large thumbs-up gesture beside the face, amplifying emotional expression.

Social Mediation During hands-free calls, *EmoMime* places a hand over the ear to signal the “on a call” social context to surroundings. It also performs a “cupping ear” gesture to show intent to listen, or conversely, the “blocking ear” gesture to indicate intent to block noise and focus.

Self-Regulation Utilizing flexible arm compliance, *EmoMime* performs safe “self-touch” gestures. Through actions like supporting the chin while thinking or gently touching the cheek, *EmoMime* supports the user’s internal state and physically externalizes the thought process.

In this way, *EmoMime* proposes a paradigm shift in wearable robotics: from a mere “tool for tasks” to a “Social Cyborg” that extends the user’s emotional and social presence.

Acknowledgments

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